

National University of Computer and Emerging Sciences, Lahore Campus



Course Name:	Digital Signal Processing	Course Code:	EE302
Program:	BS(Electrical Engineering)	Semester:	Spring 2017
Duration:	180 Minutes	Total Marks:	100
Paper Date:	17 th May 2017	Weight	50
Section:	ALL	Page(s):	2
Exam Type:	Final		

Student : Name: _____ Roll No. _____ Section: _____

Instruction/Notes: Please write legibly, as readability and understandability are important during marking. In case of more than one attempts to a question, only one will be evaluated, whichever is found first.

Question # 1

CLO-01 [5+5=10 Points]

An analog signal contains frequencies up to 20 KHz

- What range of sampling frequencies allows exact reconstruction of this signal from its samples?
- If we sample this signal with a sampling frequency $F_s = 8$ KHz, examine what happens to frequency $F = 7$ KHz.

Question # 2

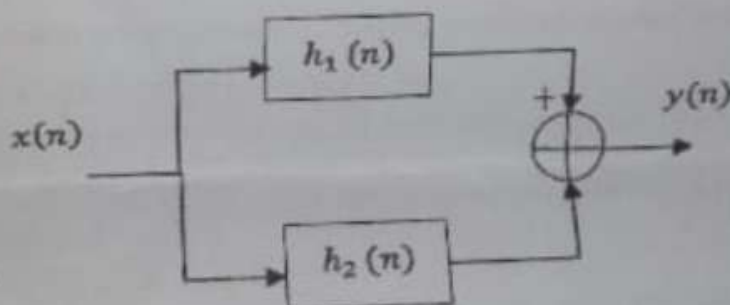
CLO-03 [15 Points]

- Consider the interconnection of LTI system. Find $y(n)$ where

$$x(n] = \{3, 1, 0, -1\}$$

$$h_1(n] = \left\{ \frac{1}{2}, \frac{1}{4}, \frac{1}{2} \right\}$$

$$h_2(n] = \{-2, 1, 0\}$$



CLO-06 [5+5+5+3=18 Points]

Question # 3

Consider the filter

$$y(n] = 0.9y[n-1] + bx[n]$$

- Determine b such that the frequency response of the filter satisfies the condition: $|H(0)] = 1$
- Draw pole zero plot of the system function $H(z)$
- Determine the frequency at which $|H(\omega)] = \frac{1}{\sqrt{2}}$
- Would you classify this filter as low pass, high pass, or bandpass? Is this filter stable? Justify your answer.

Question # 4

CLO-01[12+5=17 Points]

The values of DTFT of sequence $x(n) = \{a, b, c\}$ for $0 \leq n \leq 2$, at frequencies $\omega = \frac{\pi}{2}$, 3π and 6π are given by $3-j$, 0 , and 2 respectively.

- Determine the values of samples a, b, c .
- What is the advantage of Discrete Time Fourier Series over Continuous Time Fourier Series?

Question # 5

CLO-02 [10+10=20 Points]

Consider the sequence $x(n) = 5\delta(n) + 2\delta(n-2) + 3\delta(n-3)$. If $X(k)$ is the 4 point DFT of $x(n)$

- Find the finite sequence $y_1(n)$ if $Y_1[k] = W_4^{-3k}X[k]$.
- The four-point DFT of $x(n)$ is computed and the resulting sequence is squared such that $Y_2[k] = X^2[k]$. A four-point inverse DFT is then computed to produce sequence $y_2(n)$. Find the sequence $y_2(n)$.

Question # 6

[5+10+5=20 Points]

- Find all possible $x(n)$ associated with the Z-transform:

$$X(z) = \frac{5z^{-1}}{(3-z^{-1})(1-2z^{-1})}$$

- A causal linear shift-invariant system is characterized by the difference equation

$$y(n) = \frac{1}{4}y(n-1) + \frac{1}{8}y(n-2) + x(n) - x(n-1)$$

Find the system function $H(z)$, and the unit sample response $h(n)$, and determine whether or not system is stable.

- The z-transform of a sequence $x(n)$ is:

$$X(z) = \frac{1-4z^{-1}+2z^{-2}}{1-3z^{-1}+0.5z^{-2}}$$

Find the DTFT of $x(n)$ at $\omega = \pi/2$

Formulae:

$$\text{DFT: } X(k) = \sum_{n=0}^{N-1} x(n)e^{-j\frac{2\pi kn}{N}} \quad k = 0, 1, 2, \dots, N-1 \quad \text{IDFT: } x(n) = \frac{1}{N} \sum_{k=0}^{N-1} X(k)e^{j\frac{2\pi kn}{N}} \quad n = 0, 1, 2, \dots, N-1$$

$$\text{IDTFS: } x(n) = \sum_{k=0}^{N-1} c_k e^{j\frac{2\pi kn}{N}}$$

$$\text{DTFS: } c_k = \frac{1}{N} \sum_{n=0}^{N-1} X(k)e^{-j\frac{2\pi kn}{N}}$$

$$\text{DTFT: } X(\omega) = \sum_{n=-\infty}^{\infty} x(n)e^{-j\omega n}$$